Since f is continuous across τ, $\widehat{\sigma}_{K\tau^-} = \widehat{\sigma}_{K'\tau^+}$, where K, K' are such that they correspond to the same value of y:

$$\frac{K' + \delta K(\tau^+)}{\alpha(\tau^+)S_0 - \delta S(\tau^+)} = \frac{K + \delta K(\tau^-)}{\alpha(\tau^-)S_0 - \delta S(\tau^-)} \tag{2.27}$$

From the definition of $\alpha, \delta S, \delta K$:

$$\begin{aligned}\alpha(\tau^+) &= (1-z)\alpha(\tau^-)\\ \delta S(\tau^+) &= (1-z)\delta S(\tau^-)\\ \delta K(\tau^+) &= (1-z)\delta K(\tau^-) + c\end{aligned}$$

which, once plugged in (2.27) yields:

$$K' = (1-z)K - c$$

Thus (2.23) is exactly obeyed: using a smooth function $f(t, y)$ with y given by (2.26) automatically takes care of the matching conditions across dividend dates. Our final recipe is thus:

- Build a smooth interpolation of $f = (t - t_0)\,\widehat{\sigma}^2_{Kt}$ as a function of (t, y) with y defined in (2.26).
- Use formula (2.19) to generate the local volatility function.

In the author's experience this approximate technique is accurate for indexes (many small dividends) and stocks (few large dividends) alike – see Figure 2.1 for an example with the S&P 500 index. As an additional benefit, whenever we input a flat volatility surface – $\widehat{\sigma}_{KT} = \sigma_0, \forall\, K, T$ – we exactly recover a flat local volatility function: $\sigma(t, S) = \sigma_0, \forall\, t, S$.[6]

2.4 From local volatilities to implied volatilities

Expression (2.19) gives local volatilities as a function of implied volatilities. For the sake of analyzing the dynamics of the local volatility model, we need to study how, for a set local volatility function $\sigma(t, S)$, implied volatilities respond to a move of S. Rather than solving the forward equation (2.7) for call option prices, we will use an approximate formula that expresses implied volatilities as a function of the local volatility function directly.

We first derive a more general identity.

[6] The ratios $\frac{T-t_i}{T}$, $\frac{t_i}{T}$ in the definition of δS, δK, could be replaced by other functions of $\frac{t_i}{T}$, provided these vanish respectively for $t_i = T$ and $t_i = 0$, for the sake of ensuring condition (2.23). We leave this optimization to the reader.

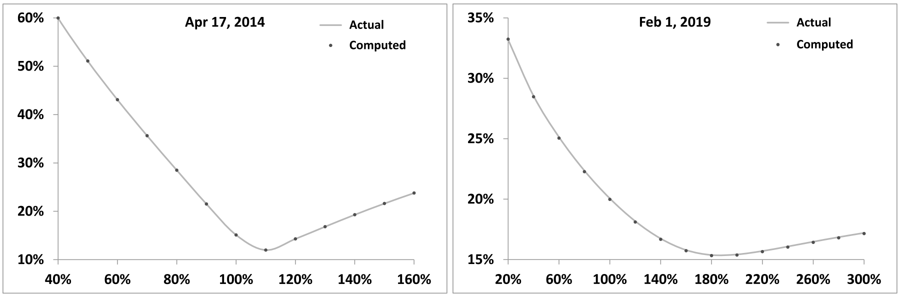

Figure 2.1: Comparison of market implied volatilities (solid line) with implied volatilities computed using the local volatility function generated by the approximate technique of Section 2.3.1.2 (dots), for the S&P 500 index, for two maturities. Market parameters as of February 1, 2014 have been used.

2.4.1 Implied volatilities as weighted averages of instantaneous volatilities

Consider a base model – denoted model I – a local volatility model with volatility function $\sigma_1(t,S)$. Denote by $P_1(t,S)$ the price of a vanilla option of strike K, maturity T in this base model.

Consider another model – denoted model II – such that the dynamics of S_t is given by:

$$dS_t = (r-q)\, S_t dt + \sigma_{2t} S_t dW_t \tag{2.28}$$

where the instantaneous volatility σ_{2t} is for now an arbitrary process.

Imagine delta-hedging a short vanilla option position of maturity T using the base model with the actual dynamics of S_t given by the second model. Our final P&L is the sum of all gamma/theta P&Ls between successive delta rehedges, each one equal to: $-S_t^2 \left.\frac{d^2P_1}{dS^2}\right|_{S=S_t} \left(\sigma_{2t}^2 - \sigma_1(t,S_t)^2\right) dt$.

The price P_2 we should charge for this option is then the price in model I plus the opposite of an estimate of the sum of such gamma/theta P&Ls. The following derivation makes this trading intuition more precise.

Consider process Q_t defined by:

$$Q_t = e^{-rt} P_1(t,S_t)$$

At $t=0$, $Q_{t=0}$ is simply the initial price in model I, for the initial spot value S_0. The variation during dt of Q_t reads, in the dynamics of model II:

$$\begin{aligned} dQ_t &= e^{-rt}\left[\Big(-rP_1+\frac{dP_1}{dt}\Big)dt+\frac{dP_1}{dS}dS_t+\frac{1}{2}\frac{d^2P_1}{dS^2}\left\langle dS_t^2\right\rangle\right] \\ &= e^{-rt}\left[\Big(-rP_1+\frac{dP_1}{dt}\Big)dt+\frac{dP_1}{dS}dS_t+\frac{1}{2}\sigma_{2t}^2S_t^2\frac{d^2P_1}{dS^2}dt\right] \end{aligned} \tag{2.29}$$

where P_1 and its derivatives are evaluated at (t,S_t). Taking the expectation of (2.29) yields:

$$\begin{aligned} E_2[dQ_t|t,S_t] &= e^{-rt}\Big(-rP_1+\frac{dP_1}{dt}+(r-q)\,S_t\frac{dP_1}{dS}+\frac{1}{2}\sigma_{2t}^2S_t^2\frac{d^2P_1}{dS^2}\Big)dt \\ &= e^{-rt}\frac{S_t^2}{2}\frac{d^2P_1}{dS^2}\left(\sigma_{2t}^2-\sigma_1(t,S_t)^2\right)dt \end{aligned}$$

where we have made use of pricing equation (2.2) for P_1 and where the subscript 2 indicates that the expectation is taken over paths of S_t generated by SDE (2.28). Integrating the above expression on $[0,T]$:

$$\begin{aligned} E_2[Q_T] &= Q_0+\int_0^T E_2[dQ_t] \\ &= P_1\left(0,S_0\right)+E_2\left[\int_0^T e^{-rt}\frac{S_t^2}{2}\frac{d^2P_1}{dS^2}\left(\sigma_{2t}^2-\sigma_1(t,S_t)^2\right)dt\right] \end{aligned}$$

Now $Q_T=e^{-rT}P_1\left(T,S_T\right)=e^{-rT}f\left(S_T\right)$, where f is the European option's payoff, thus $E_2[Q_T]=e^{-rT}E_2[f\left(S_T\right)]=P_2\left(0,S_0,\bullet\right)$ where P_2 is the pricing function of model 2, which involves its own parameters in addition to t,S. We thus have:

$$P_2\left(0,S_0,\bullet\right) = P_1\left(0,S_0\right)+E_2\left[\int_0^T e^{-rt}\frac{S_t^2}{2}\frac{d^2P_1}{dS^2}\left(\sigma_{2t}^2-\sigma_1(t,S_t)^2\right)dt\right] \tag{2.30}$$

which expresses mathematically what trading intuition suggested.

The price of an option for an arbitrary dynamics of S_t is given by its price in a given base model – here a local volatility model – supplemented with the expectation of the discounted gamma/theta P&Ls incurred as one delta-hedges the option using the base model until maturity.

Assume now that model I is the Black-Scholes model with a constant volatility equal to the implied volatility of the vanilla option at hand, backed out of model II price: $\sigma_1(t,S_t)\equiv\widehat{\sigma}_{KT}$. By definition of $\widehat{\sigma}_{KT}$:

$$P_1\left(0,S_0\right)=P_{\widehat{\sigma}_{KT}}\left(0,S_0\right)=P_2\left(0,S_0,\bullet\right)$$

where $P_{\widehat{\sigma}_{KT}}\left(t,S\right)$ denotes the Black-Scholes price with volatility $\widehat{\sigma}_{KT}$. (2.30) then yields:

$$\widehat{\sigma}_{KT}^2 = \frac{E_2\left[\int_0^T e^{-rt} S_t^2 \frac{d^2 P_{\widehat{\sigma}_{KT}}}{dS^2} \sigma_{2t}^2 dt\right]}{E_2\left[\int_0^T e^{-rt} S_t^2 \frac{d^2 P_{\widehat{\sigma}_{KT}}}{dS^2} dt\right]} \tag{2.31}$$

This is true for an arbitrary instantaneous volatility σ_{2t}. Specialize now to the case of a local volatility model: $\sigma_{2t} \equiv \sigma(t, S)$ and $\widehat{\sigma}_{KT}$ is the implied volatility corresponding to the local volatility function. We have:

$$\widehat{\sigma}_{KT}^2 = \frac{E_{\sigma(t,S)}\left[\int_0^T e^{-rt} S_t^2 \frac{d^2 P_{\widehat{\sigma}_{KT}}}{dS^2} \sigma\left(t, S\right)^2 dt\right]}{E_{\sigma(t,S)}\left[\int_0^T e^{-rt} S_t^2 \frac{d^2 P_{\widehat{\sigma}_{KT}}}{dS^2} dt\right]} \tag{2.32}$$

$\widehat{\sigma}_{KT}^2$ is thus the average value of $\sigma\left(t, S\right)^2$, weighted by the dollar gamma computed with the constant volatility $\widehat{\sigma}_{KT}$ itself, over paths generated by the local volatility $\sigma\left(t, S\right)$.

Going back to identity (2.30), consider instead that model I is the local volatility model – $\sigma_1\left(t, S\right) \equiv \sigma\left(t, S\right)$ – and that model II is the Black-Scholes model with volatility $\widehat{\sigma}_{KT}$. Again we have $P_1\left(0, S_0\right) = P_2\left(0, S_0\right)$, and now get:

$$\widehat{\sigma}_{KT}^2 = \frac{E_{\widehat{\sigma}_{KT}}\left[\int_0^T e^{-rt} S^2 \frac{d^2 P_{\sigma(S,t)}}{dS^2} \sigma\left(t, S\right)^2 \ dt\right]}{E_{\widehat{\sigma}_{KT}}\left[\int_0^T e^{-rt} \ S^2 \frac{d^2 P_{\sigma(S,t)}}{dS^2} dt\right]} \tag{2.33}$$

where the averaging is now performed using the density generated by a Black-Scholes model with constant volatility $\widehat{\sigma}_{KT}$ and $\sigma\left(t, S\right)^2$ is now weighted by the dollar gamma computed in the local volatility model.[7]

In this derivation we have used as base model the local volatility model, with a fixed volatility function – or the Black-Scholes model, a particular version of it – and have considered a delta-hedged vanilla option position.

In Section 8.4, page 316, we derive a different expression for European option prices in diffusive models by considering a delta-hedged, vega-hedged option position, using as base model a Black-Scholes model whose volatility is constantly recalibrated to the terminal VS volatility.

2.4.2 Approximate expression for weakly local volatilities

While the Dupire formula (2.19) explicitly expresses local volatilities as a function of implied volatilities, formulas (2.32) and (2.33) for $\widehat{\sigma}_{KT}$ are implicit, as the right-hand side depends on the unknown implied volatility $\widehat{\sigma}_{KT}$, either through the dollar gamma or through the density used for averaging the numerator and denominator.

[7] As far as I can remember, expression (2.32) for implied volatilities in the local volatility model was first presented by Bruno Dupire at a Global Derivatives conference in the late 90s.

In order to get a more usable expression, we will assume that $\sigma(t,S)$ is only weakly local.

We will use formula (2.32) with the Black-Scholes model with time-dependent volatility $\sigma_0(t)$ as base model.

Let us use the local variance $u(t,S) = \sigma(t,S)^2$ and assume that

$$u(t,S) \;=\; u_0(t) + \delta u(t,S)$$

where $u_0 = \sigma_0^2(t)$ and δu is a small perturbation. If $\delta u = 0$, $\widehat{\sigma}_{KT} = \widehat{\sigma}_{0T}$ where $\widehat{\sigma}_{t_1 t_2}$ is defined as:

$$\widehat{\sigma}^2_{t_1 t_2} \;=\; \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} \sigma_0^2(t)dt$$

Let us use expression (2.32) and expand $\widehat{\sigma}^2_{KT}$ at first order in δu: $\widehat{\sigma}^2_{KT} + \delta(\widehat{\sigma}^2_{KT})$. $u(t,S)$ appears explicitly in the numerator as well as implicitly in the density used for computing expectations in both numerator and denominator. For the sake of computing the order-one perturbation in δu, however, the contribution generated by the density vanishes: from equation (2.32) and using a compact notation:

$$\begin{aligned}\widehat{\sigma}^2_{KT} + \delta(\widehat{\sigma}^2_{KT}) \;&=\; \frac{E_{u_0+\delta u}[(u_0+\delta u)\,\bullet]}{E_{u_0+\delta u}[\bullet]} \;=\; u_0 + \frac{E_{u_0+\delta u}[\delta u\;\bullet]}{E_{u_0+\delta u}[\bullet]}\\ &=\; u_0 + \frac{E_{u_0}[\delta u\;\bullet]}{E_{u_0}[\bullet]}\end{aligned} \tag{2.34}$$

where $\bullet$ is computed at order zero in δu. We thus have:

$$\delta(\widehat{\sigma}^2_{KT}) \;=\; \frac{E_{\sigma_0}\left[\int_0^T e^{-rt}\,\delta u(t,S)\,S^2\frac{d^2P_{\sigma_0}}{dS^2}dt\right]}{E_{\sigma_0}\left[\int_0^T e^{-rt}\;S^2\frac{d^2P_{\sigma_0}}{dS^2}dt\right]} \tag{2.35}$$

The right-hand side of equation (2.35) now only requires the density and the dollar gamma of a call or put option, evaluated in the Black-Scholes model with deterministic volatility $\sigma_0(t)$ – both analytically known. The denominator in (2.35) involves the discounted dollar gamma of a European option, averaged over its lifetime. In the Black-Scholes model, $e^{-rt}\;S^2\frac{d^2P_{\sigma_0}}{dS^2}$ is a martingale – see a proof in Appendix A of Chapter 5, page 181. Thus:

$$E_{\sigma_0}\left[e^{-rt}\;S^2\frac{d^2P_{\sigma_0}}{dS^2}\right] \;=\; S_0^2\;\frac{d^2P_{\sigma_0}}{dS^2}\bigg|_{t=0,S=S_0} \tag{2.36}$$

where S_0 denotes today's spot price. The denominator is then equal to $T\;S_0^2\;\frac{d^2P_{\sigma_0}}{dS^2}\Big|_{t=0,\;S=S_0}$.

Focus now on the numerator in (2.35). It reads:

$$\int_0^T dt\int_0^\infty dS\;\rho_{\sigma_0}(t,S)\,e^{-rt}\;\delta u(t,S)\;\;S^2\frac{d^2P_{\sigma_0}}{dS^2} \tag{2.37}$$

where $\rho_{\sigma_0}(t,S)$ is the lognormal density with deterministic volatility $\sigma_0(t)$. Define $x = \ln(S/F_t)$ where F_t is the forward for maturity t: $F_t = S_0 e^{(r-q)t}$. $\rho_{\sigma_0}(t,S)$ and $S^2 \frac{d^2 P_{\sigma_0}}{dS^2}$ are given by:

$$\rho_{\sigma_0}(t,S) = \frac{1}{\sqrt{2\pi\omega_t}S} e^{-\frac{\left(x+\frac{\omega_t}{2}\right)^2}{2\omega_t}} \tag{2.38}$$

$$S^2 \frac{d^2 P_{\sigma_0}}{dS^2} = S\frac{F_T}{F_t} e^{-r(T-t)} \frac{1}{\sqrt{2\pi(\omega_T-\omega_t)}} e^{-\frac{\left(-x_K+x+\frac{(\omega_T-\omega_t)}{2}\right)^2}{2(\omega_T-\omega_t)}} \tag{2.39}$$

where $x_K = \ln(\frac{K}{F_T})$ and we have introduced the integrated variance ω_t, defined as:

$$\omega_t = \int_0^t \sigma_0^2(\tau)d\tau$$

The numerator then reads:

$$F_T e^{-rT} \int_0^T dt \int_{-\infty}^{+\infty} dx \, \frac{\delta u\,(t, F_t e^x)}{\sqrt{2\pi(\omega_T-\omega_t)}\sqrt{2\pi\omega_t}} e^x e^{-\frac{\left(-x_K+x+\frac{(\omega_T-\omega_t)}{2}\right)^2}{2(\omega_T-\omega_t)}} e^{-\frac{\left(x+\frac{\omega_t}{2}\right)^2}{2\omega_t}}$$

Combining the exponentials, we can rewrite this expression as:

$$F_T e^{-rT} \int_0^T dt \int_{-\infty}^{+\infty} dx \frac{\delta u\,(t, F_t e^x)}{\sqrt{2\pi(\omega_T-\omega_t)}\sqrt{2\pi\omega_t}} e^{-\left(\frac{1}{2(\omega_T-\omega_t)}+\frac{1}{2\omega_t}\right)\frac{(\omega_T x - \omega_t x_K)^2}{\omega_T^2}} e^{-\frac{\left(x_K-\frac{\omega_T}{2}\right)^2}{2\omega_T}}$$

We now divide this by (2.36), the dollar gamma evaluated at $t=0$ and multiplied by T, which reads:

$$T F_T e^{-rT} \frac{1}{\sqrt{2\pi\omega_T}} e^{-\frac{\left(-x_K+\frac{\omega_T}{2}\right)^2}{2\omega_T}}$$

and get for the ratio in (2.35):

$$\frac{1}{T}\int_0^T dt \int_{-\infty}^{+\infty} dx \, \frac{\sqrt{\omega_T}}{\sqrt{\omega_t}} \frac{\delta u\,(t, F_t e^x)}{\sqrt{2\pi(\omega_T-\omega_t)}} e^{-\left(\frac{1}{2(\omega_T-\omega_t)}+\frac{1}{2\omega_t}\right)\frac{(\omega_T x - \omega_t x_K)^2}{\omega_T^2}}$$

Switching now from x to a new coordinate y:

$$x = \frac{\omega_t}{\omega_T} x_K + \frac{\sqrt{(\omega_T-\omega_t)\omega_t}}{\sqrt{\omega_T}} y$$

yields our final formula for $\delta(\widehat{\sigma}_{KT}^2)$ at order one in δu:

$$\delta(\widehat{\sigma}_{KT}^2) = \frac{1}{T}\int_0^T dt \int_{-\infty}^{+\infty} dy \, \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \, \delta u\left(t, F_t e^{\frac{\omega_t}{\omega_T} x_K + \frac{\sqrt{(\omega_T-\omega_t)\omega_t}}{\sqrt{\omega_T}} y}\right)$$

Noting that, at order zero, $\widehat{\sigma}_{KT}^2 = \frac{1}{T}\int_0^T \sigma_0^2(t)dt = \frac{1}{T}\int_0^T u_0(t)dt$, this can be rewritten, at order one in $\delta u = \sigma^2(t,S) - \sigma_0^2(t)$, as:

$$\widehat{\sigma}_{KT}^2 \;=\; \frac{1}{T}\int_0^T dt \int_{-\infty}^{+\infty} dy\, \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}}\, u\left(t, F_t e^{\frac{\omega_t}{\omega_T}x_K + \frac{\sqrt{(\omega_T-\omega_t)\omega_t}}{\sqrt{\omega_T}}y}\right) \tag{2.40}$$

This is the expression of $\widehat{\sigma}_{KT}$ at order one in the perturbation of σ around a time-dependent volatility $\sigma_0(t)$. The square of the implied volatility is expressed as an integral of the square of the instantaneous volatility – thus (2.40) is exact when u is a function of t only.[8]

2.4.3 Expanding around a constant volatility

Consider as base case the Black-Scholes model with constant volatility σ_0: $\sigma_0(t) = \sigma_0$, thus $\omega_t = \sigma_0^2 t$ and write:

$$\sigma(t,S) \;=\; \sigma_0 + \delta\sigma(t,S)$$

thus $\delta u = 2\sigma_0\delta\sigma(t,S)$. Using (2.40) together with $\widehat{\sigma}_{KT}^2 = \sigma_0^2 + 2\sigma_0\delta(\widehat{\sigma}_{KT})$ yields:

$$\delta\widehat{\sigma}_{KT} \;=\; \frac{1}{T}\int_0^T dt \int_{-\infty}^{+\infty} dy\, \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}}\, \delta\sigma\left(t, F_t e^{\frac{t}{T}x_K + \sigma_0\sqrt{\frac{(T-t)t}{T}}y}\right) \tag{2.41}$$

Adding σ_0 on both sides yields, at order one in $\delta\sigma(t,S)$:

$$\widehat{\sigma}_{KT} \;=\; \frac{1}{T}\int_0^T dt \int_{-\infty}^{+\infty} dy\, \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}}\, \sigma\left(t, F_t e^{\frac{t}{T}x_K + \sigma_0\sqrt{\frac{(T-t)t}{T}}y}\right) \tag{2.42}$$

Even though (2.40) may be slightly more accurate when the term-structure of volatilities is strong, we will use (2.42) in the sequel, since resulting expressions for the ATMF skew and the SSR are simpler.

2.4.4 Discussion

For $t = 0$, $F_t \exp\left(\frac{t}{T}\ln\frac{K}{F_T} + \frac{\sqrt{\sigma_0^2(T-t)t}}{\sqrt{T}}y\right) = S_0$. Thus values of $\sigma\,(t=0,S)$ for $S \neq S_0$ do not contribute in (2.42) to $\widehat{\sigma}_{KT}$.

Likewise, for $t=T$, $F_t \exp\left(\frac{t}{T}\ln\frac{K}{F_T} + \frac{\sqrt{\sigma_0^2(T-t)t}}{\sqrt{T}}y\right) = K$: values of $\sigma\,(t=T,S)$ for $S \neq K$ do not contribute either. This is natural, upon inspection of expression

[8] This is comforting but should not cause too much rejoicing – (2.40) is only an order-one approximation, besides we will see further below that for $T \to 0$ there is an exact expression of the *inverse* of $\widehat{\sigma}_{KT}$ as an average of the *inverse* of $\sigma\,(0,S)$.

(2.37): for $t = 0$, the density ρ_{σ_0} vanishes unless $S = S_0$, while for $t = T$ the dollar gamma $S^2 \frac{d^2 P_{\sigma_0}}{dS^2}$ vanishes, unless $S = K$.

In (2.42), the largest weight is obtained for $y = 0$: this singles out a path for $\ln(S)$ which is a straight line starting at $\ln(S_0)$ for $t = 0$ and ending at $\ln(K)$ at $t = T$. Replacing the integral over y by the value for $y = 0$ would give an approximation of $\widehat{\sigma}_{KT}$ as a uniform average of the local volatility along this line:

$$\widehat{\sigma}_{KT} \simeq \frac{1}{T} \int_0^T \sigma \left(t, F_t e^{\frac{t}{T} \ln \frac{K}{F_T}} \right) dt \tag{2.43}$$

Summing over values of $y \neq 0$, includes other paths in the average, with their ends pinned down at S_0 at $t = 0$ and at K for $t = T$ by the factor $\sqrt{\sigma_0^2 (T-t)\, t}$.

However appealing expression (2.42) for $\widehat{\sigma}_{KT}$ may be, market smiles on equity underlyings are strong enough, and bid-offer spreads on vanilla option prices are tight enough, that its numerical accuracy is not sufficient for practical trading purposes: an order one expansion in $\delta\sigma(t, S)$ is simply not adequate.[9]

Can we do better? Expression (2.42) for $\widehat{\sigma}_{KT}$ amounts to merely setting $\widehat{\sigma}_{KT} = \sigma_0$ and using the lognormal density with volatility σ_0 for computing both averages in the numerator and denominator of the right-hand side of equation (2.32).

A number of tricks have been proposed under the loose name of "most likely path" techniques for approximating the right-hand side of (2.32), using a lognormal density but with a different implied volatility for each time slice – still, their accuracies are not adequate.

The reason for this is that formula (2.32) seems to suggest that the main contribution of $\sigma(t, S)$ is embodied in the explicit gamma term in the numerator and that using an approximate density – for example lognormal – for computing both averages in the numerator and the denominator will do. This is not the case. In practice, taking into account the dependence of the density itself on $\sigma(t, S)$ is mandatory for achieving the accuracy needed in trading applications.[10] For realistic equity smiles, there isn't yet a computationally efficient alternative to numerically solving the forward equation (2.7).

Is formula (2.42) then of any use? While a good approximation of absolute volatility levels is hard to come by, the skew – which is the difference of two volatilities – is more easily approximated. We now use equation (2.42) to calculate the skew and curvature of the smile, as a function of parameters of the local volatility function.

[9]The cruder version (2.43) is even less usable.

[10]Julien Guyon and Pierre Henry-Labordère provide in [51] a nice summary and comparison of different "most likely path" approximations, along with a technique based on a short-time heat-kernel expansion for ρ.

2.4.5 The smile near the forward

Let us assume that the local volatility is smooth and given by:

$$\sigma(t,S) \;=\; \overline{\sigma}(t) + \alpha(t)\,x + \frac{\beta(t)}{2}x^2 \tag{2.44}$$

where x – which we call moneyness – is given by $x = \ln(S/F_t)$.

We assume that $\alpha(t)$, $\beta(t)$ are small, and that $\overline{\sigma}(t)$ does not vary too much, so that the difference $\overline{\sigma}(t) - \sigma_0$ is small, where σ_0 is the constant volatility level around which the order-one expansion in (2.35) is performed.

We could as well perform the expansion around the time-deterministic volatility $\overline{\sigma}(t)$ – calculations are similar. For the sake of simplicity we carry out the expansion around a constant σ_0 – expressions for the more general case appear in (2.60), page 51.

Equation (2.42) gives :

$$\begin{aligned}\widehat{\sigma}_{KT} \;&=\; \frac{1}{T}\int_0^T \overline{\sigma}(t)\,dt \\ &+\; \frac{1}{T}\int_0^T dt \int_{-\infty}^{+\infty} dy\, \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} \left[\alpha(t)X(t,y) + \frac{\beta(t)}{2}X(t,y)^2\right]\end{aligned} \tag{2.45}$$

where

$$X(t,y) \;=\; \frac{t}{T}x_K \;+\; \frac{\sqrt{\sigma_0^2\,(T-t)\,t}}{\sqrt{T}}y \tag{2.46}$$

Doing the integrals over y we get at order 1 in α, β:

$$\begin{aligned}\widehat{\sigma}_{KT} \;&=\; \frac{1}{T}\int_0^T \overline{\sigma}(t)\,dt \;+\; \frac{\sigma_0^2 T}{2}\frac{1}{T}\int_0^T \frac{(T-t)\,t}{T^2}\beta(t)dt \\ &+\; \left(\frac{1}{T}\int_0^T \frac{t}{T}\alpha(t)dt\right)x_K \;+\; \frac{1}{2}\left(\frac{1}{T}\int_0^T \frac{t^2}{T^2}\beta(t)dt\right)x_K^2\end{aligned} \tag{2.47}$$

Thus, for a sufficiently smooth local volatility function, the skew and curvature of the smile near the forward are related to the skew and curvature of the local volatility function through:

$$\mathcal{S}_T \;=\; \left.\frac{d\widehat{\sigma}_{KT}}{d\ln K}\right|_{F_T} \;=\; \frac{1}{T}\int_0^T \frac{t}{T}\alpha(t)dt \tag{2.48}$$

$$\left.\frac{d^2\widehat{\sigma}_{KT}}{d\ln K^2}\right|_{F_T} \;=\; \frac{1}{T}\int_0^T \left(\frac{t}{T}\right)^2 \beta(t)dt \tag{2.49}$$

where we have introduced $\mathcal{S}_T$ as a notation for the ATMF (at the money forward) skew.[11]

[11] These approximate formulas for the implied ATMF skew and curvature in the local volatility model can be obtained in a number of ways – see [78] for an alternative derivation of the “skew-averaging” expression (2.48).

2.4.5.1 A constant local volatility function

Assume that α and β are constant. We get:

$$\left.\frac{d\widehat{\sigma}_{KT}}{d\ln K}\right|_{F_T} = \frac{1}{T}\int_0^T \frac{t}{T}dt\,\alpha = \frac{\alpha}{2} \tag{2.50a}$$

$$\left.\frac{d^2\widehat{\sigma}_{KT}}{d\ln K^2}\right|_{F_T} = \frac{1}{T}\int_0^T \left(\frac{t}{T}\right)^2 \beta dt = \frac{\beta}{3} \tag{2.50b}$$

Thus, for a local volatility function of the form (2.44) with α and β constant, at order one in α, β, the ATMF skew of the implied volatility is half the skew of the local volatility function, while its curvature is one third the curvature of the local volatility function.

2.4.5.2 A power-law-decaying ATMF skew

Let us assume a power-law form for $\alpha(t)$. To prevent divergence as $t \to 0$ we take:

$$\begin{cases} \alpha(t) = \alpha_0 \left(\frac{\tau_0}{t}\right)^\gamma & t > \tau_0 \\ \alpha(t) = \alpha_0 & t \leq \tau_0 \end{cases} \tag{2.51}$$

where τ_0 is a cutoff – typically $\tau_0 = 3$ months – and γ is the characteristic exponent of the long-term decay of $\alpha(t)$. We get, from (2.48):

$$\begin{cases} \mathcal{S}_T = \frac{\alpha_0}{2} & T \leq \tau_0 \\ \mathcal{S}_T = \frac{1}{2-\gamma}\alpha_0 \left(\frac{\tau_0}{T}\right)^\gamma - \frac{\gamma}{2(2-\gamma)}\alpha_0 \left(\frac{\tau_0}{T}\right)^2 & T \geq \tau_0 \end{cases} \tag{2.52}$$

γ is typically smaller than 2. For (very) long maturities the second piece in (2.52) can be ignored and we get:

$$\mathcal{S}_T \simeq \frac{1}{2-\gamma}\alpha_0 \left(\frac{\tau_0}{T}\right)^\gamma \tag{2.53}$$

The long-term ATMF skew thus decays with the same exponent γ as the local volatility function. For typical equity smiles, $\gamma \simeq \frac{1}{2}$ – see examples in Figure 2.3, page 58. With respect to the local volatility skew $\alpha(t)$, $\mathcal{S}_T$ is rescaled by a factor $\frac{1}{2-\gamma}$.

2.4.6 An exact result for short maturities

We consider here the case $T \to 0$, for which a particularly simple relationship linking local and implied volatilities exists, which we now derive.

Let us recall the definition of y and f, which appear in expression (2.19):

$$y = \ln\left(\frac{K}{F_T}\right)$$
$$f(T,\ y) = (T - t_0)\,\widehat{\sigma}^2_{K=F_T e^y, T}$$

To lighten the notation we use $\widehat{\sigma}(T, y)$ instead of $\widehat{\sigma}_{K=F_T e^y, T}$ and take $t_0 = 0$. Expression (2.19) reads:

$$\sigma^2 = \frac{\widehat{\sigma}^2 + 2T\widehat{\sigma}\widehat{\sigma}_T}{\left(\frac{y}{\widehat{\sigma}^2}\widehat{\sigma}\widehat{\sigma}_y - 1\right)^2 + T\left(\widehat{\sigma}_y^2 + \widehat{\sigma}\widehat{\sigma}_{yy}\right) - \left(\frac{1}{4} + \frac{1}{T\widehat{\sigma}^2}\right)\widehat{\sigma}^2\widehat{\sigma}_y^2 T^2}$$

where $\widehat{\sigma}_y, \widehat{\sigma}_T$ denote derivatives of $\widehat{\sigma}$ with respect to y and T. Take the limit $T \to 0$ and keep the leading term in the numerator and denominator in an expansion in powers of T, assuming that $\widehat{\sigma}$ is a smooth function of T and y as $T \to 0$. We get:

$$\sigma^2 = \frac{\widehat{\sigma}^2}{\left(\frac{y}{\widehat{\sigma}}\widehat{\sigma}_y - 1\right)^2} = \frac{1}{\left(\frac{y}{\widehat{\sigma}^2}\widehat{\sigma}_y - \frac{1}{\widehat{\sigma}}\right)^2}$$

which yields

$$\frac{1}{\sigma} = \pm\left(\frac{y}{\widehat{\sigma}^2}\widehat{\sigma}_y - \frac{1}{\widehat{\sigma}}\right) = \mp\left(y\left(\frac{1}{\widehat{\sigma}}\right)_y + \frac{1}{\widehat{\sigma}}\right) = \mp\left(\frac{y}{\widehat{\sigma}}\right)_y$$

Thus

$$\int_0^y \frac{du}{\sigma(T=0,\ Se^u)} = \mp\frac{y}{\widehat{\sigma}(T=0, Se^y)}$$

Following the usual convention of using positive volatilities:

$$\frac{1}{\widehat{\sigma}(T=0, Se^y)} = \frac{1}{y}\int_0^y \frac{du}{\sigma(T=0,\ Se^u)} \tag{2.54}$$

or, equivalently:

$$\frac{1}{\widehat{\sigma}(T=0, K)} = \frac{1}{\ln\frac{K}{S}}\int_S^K \frac{1}{\sigma(T=0,\ \mathcal{S})}\frac{d\mathcal{S}}{\mathcal{S}}$$

This result was first published by Henri Berestycki, Jérôme Busca and Igor Florent [6]. The fact that the inverse of $\widehat{\sigma}$ should be given by the average of the inverse of σ may surprise at first.

The squared volatility that one may have expected appears usually in averages only because they are temporal averages, akin to a quadratic variation. In our case $T \to 0$ and there is no temporal averaging.

Then note that the harmonic average complies with the basic requirement that if the local volatility vanishes in a region between the initial spot level and the strike, the implied volatility for that strike should vanish, as for $T \mapsto 0$, the effect of the drift is immaterial and the spot would be unable to cross that region.[12]

[12]The motivation for the harmonic average is that it appears naturally in the density of $\ln(S_T)$ for short maturities as the change of variable $S \to z = \int_{S_0}^S \frac{dS}{S\sigma(S)}$ results in a process for z_t which is Gaussian at short times. A derivation of (2.54) using the zeroth order of the heat-kernel expansion can be found in Section 5.2.2 of [56].